

Python-SQL-Project

Objective

The primary goal of this project was to bridge the gap between raw data storage and strategic decision-making. By integrating SQL for data extraction and Python for advanced analytics, I developed a comprehensive pipeline to uncover trends and patterns within complex datasets.

Key Components of My Work

- Data Orchestration: Managed large-scale datasets using SQL to perform complex joins, aggregations, and subqueries.
- Advanced Analytics: Leveraged Python (Pandas/NumPy) to clean, manipulate, and perform statistical analysis on the extracted data.
- Visual Intelligence: Created dynamic Data Visualizations (using Matplotlib/Seaborn) to translate technical findings into intuitive, easy-to-understand charts.

Basic Level Questions

Q1. List all unique cities where customers are located.

Python Code:

```
unique_cities = customers['customer_city'].unique()  
print(f"Total Unique Cities: {len(unique_cities)}")
```

Total Unique Cities: 4119

SQL Query:

```
SELECT COUNT(DISTINCT customer_city) AS total_unique_cities  
FROM customers;
```

Result Grid			 Filter Rows
	total_unique_cities		
▶	4119		

Q2. Count the number of orders placed in 2017.

Python Code:

```
orders_2017 = orders[orders['order_purchase_timestamp'].dt.year == 2017].shape[0]
print(f"Orders placed in 2017: {orders_2017}")
```

Orders placed in 2017: 45101

SQL Query:

```
SELECT
    YEAR(order_purchase_timestamp) AS order_year,
    COUNT(order_id) AS total_orders
FROM orders
GROUP BY YEAR(order_purchase_timestamp);
```

	order_year	total_orders
▶	2017	45101

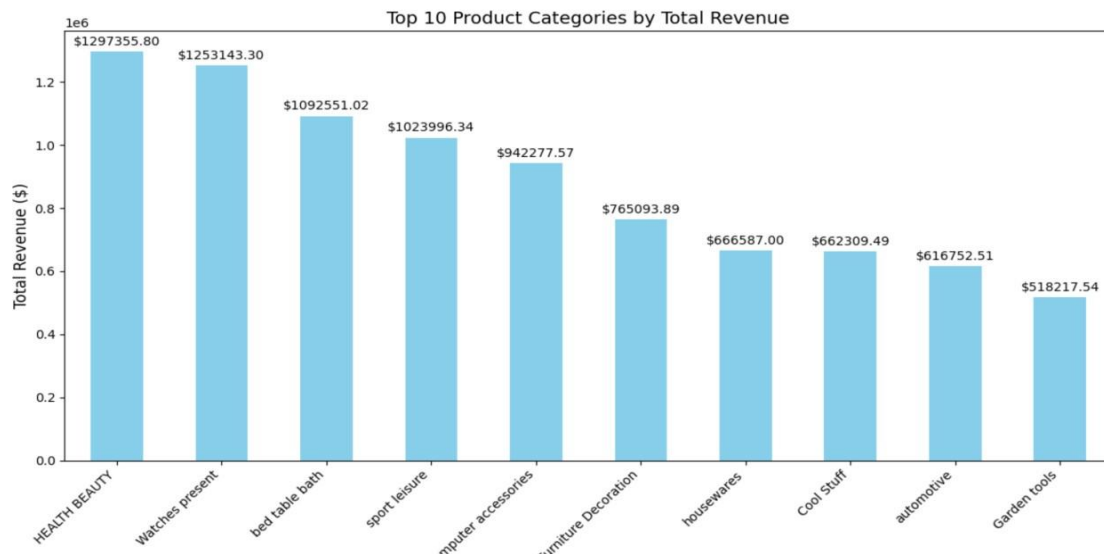
Q3. Find the total sales per category.

Python Code :

```
df = pd.merge(products, order_items, on='product_id')
df = pd.merge(df, payments, on='order_id')
top_revenue_cat = df.groupby('product category')['price'].sum().sort_values(ascending=False).head(10)

plt.figure(figsize=(12, 7))
ax = top_revenue_cat.plot(kind='bar', color='skyblue')
ax.bar_label(ax.containers[0], fmt='%.2f', padding=3)
plt.title('Top 10 Product Categories by Total Revenue', fontsize=14)
plt.xlabel('Product Category', fontsize=12)
plt.ylabel('Total Revenue ($)', fontsize=12)
plt.xticks(rotation=45, ha='right')

plt.tight_layout()
plt.show()
```



SQL Query:

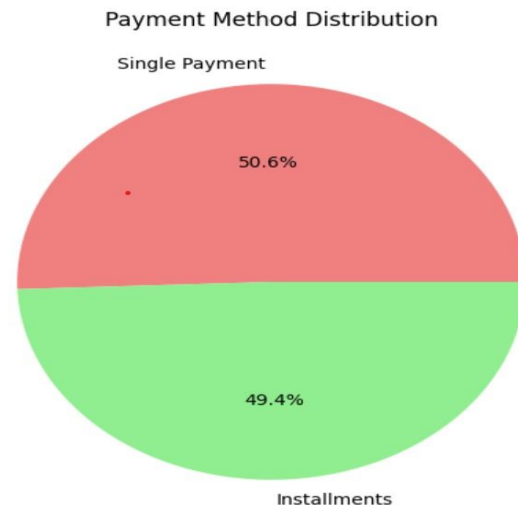
```
SELECT
    p.product_category_name,
    SUM(oi.price) AS total_revenue
FROM products p
JOIN order_items oi ON p.product_id = oi.product_id
JOIN payments pay ON oi.order_id = pay.order_id
GROUP BY p.product_category_name
ORDER BY total_revenue DESC
LIMIT 10;
```

	product_category_name	total_revenue
▶	HEALTH BEAUTY	1297355.80
	Watches present	1253143.30
	bed table bath	1092551.02
	sport leisure	1023996.34
	computer accessories	942277.57
	Furniture Decoration	765093.89
	housewares	666587.00
	Cool Stuff	662309.49
	automotive	616752.51
	Garden tools	518217.54

Q4. Calculate the percentage of orders that were paid in more than one instalment.

Python Code :

```
inst_count = payments['payment_installments'].apply(lambda x: 'Installments' if x > 1 else 'Single Payment').value_counts()
plt.figure(figsize=(6, 6))
inst_count.plot(kind='pie', autopct='%1.1f%%', colors=['lightcoral', 'lightgreen'])
plt.title('Payment Method Distribution')
plt.ylabel('')
plt.show()
```



SQL Query:

```
SELECT
    CASE
        WHEN payment_installments > 1 THEN 'Installments'
        ELSE 'Single Payment'
    END AS payment_type,
    COUNT(*) AS total_count,
    ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 1) AS percentage
FROM payments
GROUP BY 1;
```

	payment_type	total_count	percentage
▶	Installments	51338	49.4
	Single Payment	52548	50.6

5. Count the number of customers from each state.

Python Code :

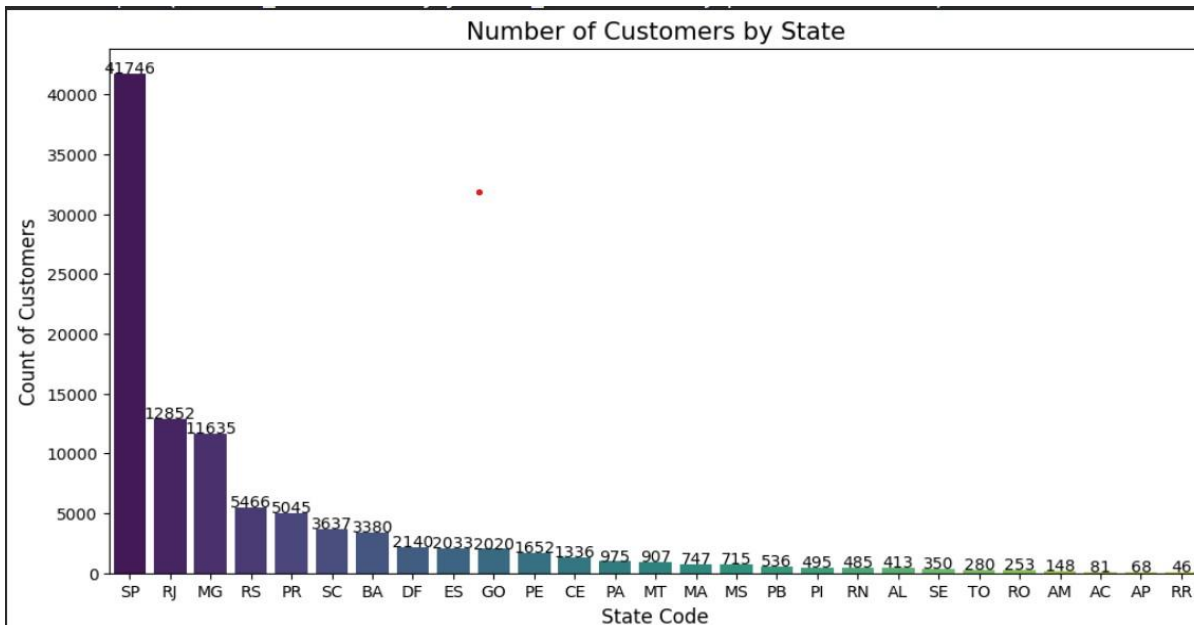
```
▶ state_counts = customers['customer_state'].value_counts()

plt.figure(figsize=(12, 6))
sns.barplot(x=state_counts.index, y=state_counts.values, palette='viridis')

plt.title('Number of Customers by State', fontsize=15)
plt.xlabel('State Code', fontsize=12)
plt.ylabel('Count of Customers', fontsize=12)

for i, value in enumerate(state_counts.values):
    plt.text(i, value + 50, str(value), ha='center', fontsize=10)

plt.show()
```



SQL Query:

```
SELECT
    customer_state,
    COUNT(customer_id) AS total_customers
FROM customers
GROUP BY customer_state
ORDER BY total_customers DESC
limit 10;
```

	customer_state	total_customers
▶	SP	41746
	RJ	12852
	MG	11635
	RS	5466
	PR	5045
	SC	3637
	BA	3380
	DF	2140
	ES	2033
	GO	2020

Intermediate Level Questions

Q1. Orders per Month in 2018.

Python Code :

```
orders_2018 = orders[orders['order_purchase_timestamp'].dt.year == 2018].copy()

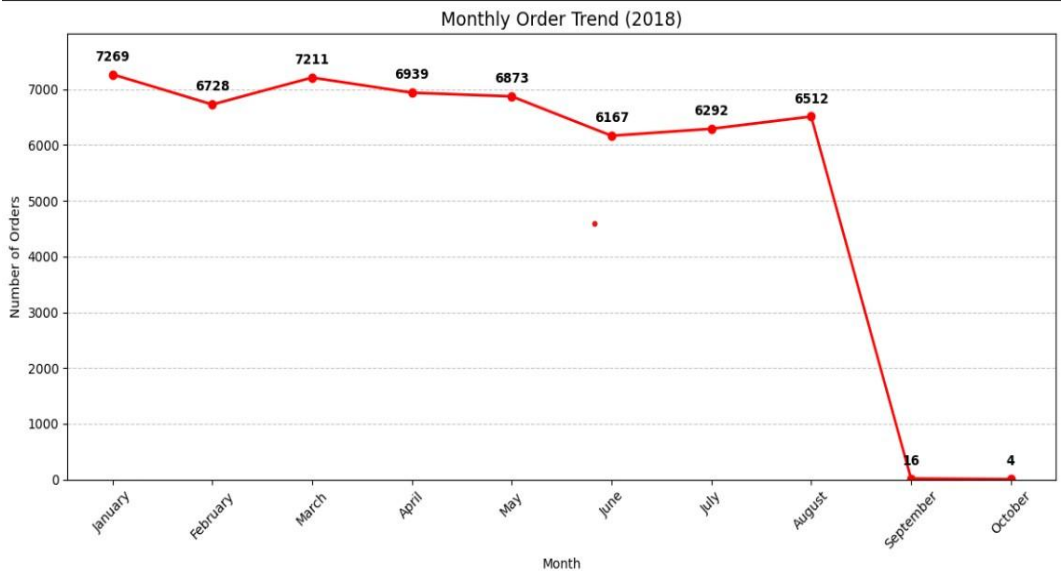
orders_2018['month_name'] = orders_2018['order_purchase_timestamp'].dt.strftime('%B')
orders_2018['month_num'] = orders_2018['order_purchase_timestamp'].dt.month

monthly_orders = orders_2018.groupby(['month_num', 'month_name']).size().reset_index(name='count')

plt.figure(figsize=(12, 6))
plt.plot(monthly_orders['month_name'], monthly_orders['count'], marker='o', color='red', linewidth=2)

for i, txt in enumerate(monthly_orders['count']):
    plt.annotate(f'{txt}', (monthly_orders['month_name'][i], monthly_orders['count'][i]),
               textcoords="offset points", xytext=(0,10), ha='center', fontsize=10, fontweight='bold')

plt.title('Monthly Order Trend (2018)', fontsize=14)
plt.ylabel('Number of Orders')
plt.xlabel('Month')
plt.xticks(rotation=45)
plt.grid(True, axis='y', linestyle='--', alpha=0.7)
plt.ylim(0, monthly_orders['count'].max() * 1.1)
plt.tight_layout()
plt.show()
```



SQL Query:

```
SELECT
    MONTHNAME(order_purchase_timestamp) AS month_name,
    COUNT(order_id) AS total_orders
FROM orders
WHERE YEAR(order_purchase_timestamp) = 2018
GROUP BY month_name
ORDER BY MIN(MONTH(order_purchase_timestamp));
```

	month_name	total_orders
▶	January	7269
	February	6728
	March	7211
	April	6939
	May	6873
	June	6167
	July	6292
	August	6512
	September	16
	October	4

Q2. Average Number of Products per Order (Grouped by City).

Python Code :

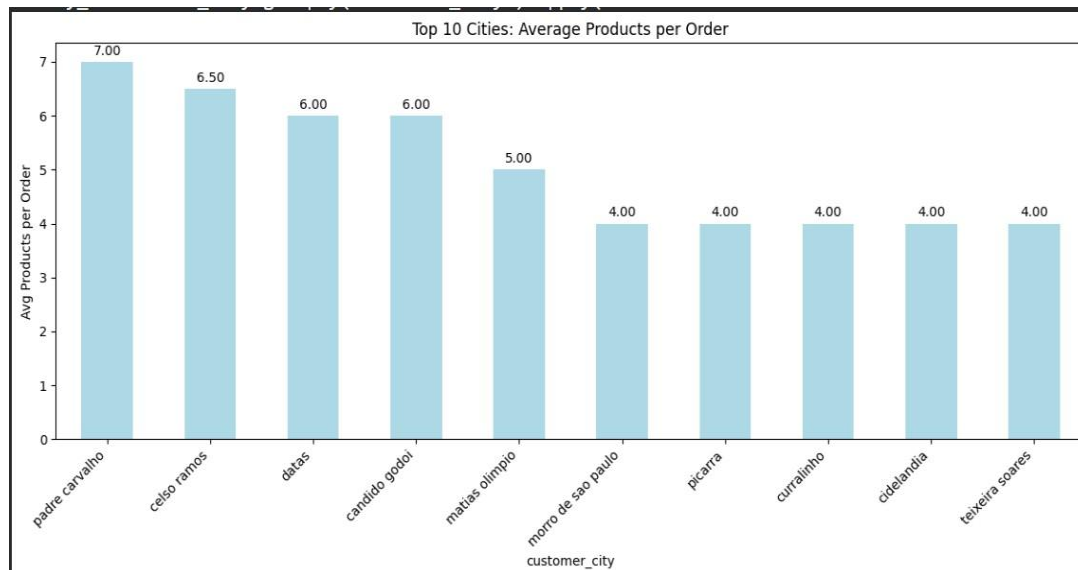
```
orders.rename(columns={orders.columns[0]: 'order_id'}, inplace=True)
df_city = pd.merge(customers, orders, on='customer_id')
df_city = pd.merge(df_city, order_items, on='order_id')

city_stats = df_city.groupby('customer_city').apply(
    lambda x: x['product_id'].count() / x['order_id'].nunique()
).sort_values(ascending=False).head(10)

plt.figure(figsize=(12, 6))
ax = city_stats.plot(kind='bar', color='lightblue')

ax.bar_label(ax.containers[0], fmt='%.2f', padding=3)

plt.title('Top 10 Cities: Average Products per Order')
plt.ylabel('Avg Products per Order')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



SQL Query:

```
SELECT
    c.customer_city,
    ROUND(COUNT(oi.product_id) / COUNT(DISTINCT oi.order_id), 2) AS avg_products_per_order
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
JOIN order_items oi ON o.order_id = oi.order_id
GROUP BY c.customer_city
ORDER BY avg_products_per_order DESC
limit 10;
```

	customer_city	avg_products_per_order
▶	padre carvalho	7.00
	celso ramos	6.50
	candido godoi	6.00
	datas	6.00
	matias olimpio	5.00
	picarra	4.00
	cidelandia	4.00
	morro de sao paulo	4.00
	teixeira soares	4.00
	curralinho	4.00

Q3. Percentage of Total Revenue by Product Category.

Python Code :

```
rev_per_cat = pd.merge(order_items[['product_id', 'price']],
                        products[['product_id', 'product category']], on='product_id')

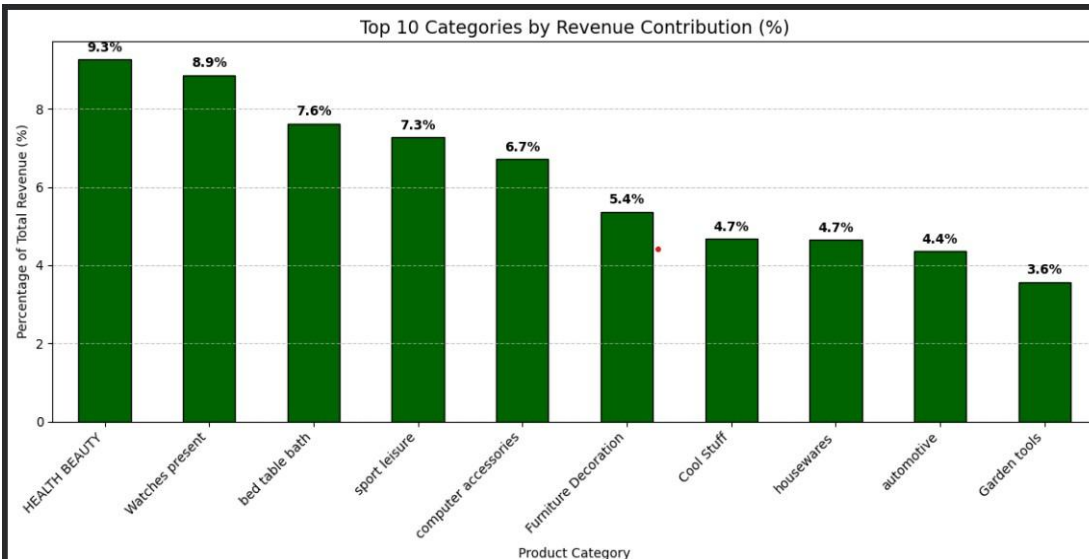
total_revenue = rev_per_cat['price'].sum()
category_revenue = rev_per_cat.groupby('product category')['price'].sum()

cat_rev_pct = (category_revenue / total_revenue) * 100
cat_rev_pct = cat_rev_pct.sort_values(ascending=False).head(10)

plt.figure(figsize=(12, 6))
ax = cat_rev_pct.plot(kind='bar', color='darkgreen', edgecolor='black')

for p in ax.patches:
    ax.annotate(f'{p.get_height():.1f}%',
                (p.get_x() + p.get_width() / 2., p.get_height()),
                ha='center', va='center',
                xytext=(0, 9),
                textcoords='offset points',
                fontsize=10, fontweight='bold')

plt.title('Top 10 Categories by Revenue Contribution (%)', fontsize=14)
plt.ylabel('Percentage of Total Revenue (%)')
plt.xlabel('Product Category')
plt.xticks(rotation=45, ha='right')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```



SQL Query:

```
SELECT
    p.product_category_name,
    ROUND(
        SUM(oi.price) * 100.0 / SUM(SUM(oi.price)) OVER(),
        1
    ) AS revenue_percentage
FROM products p
JOIN order_items oi ON p.product_id = oi.product_id
GROUP BY p.product_category_name
ORDER BY revenue_percentage DESC
LIMIT 10;
```

	product_category_name	revenue_percentage
▶	HEALTH BEAUTY	9.3
	Watches present	8.9
	bed table bath	7.6
	sport leisure	7.3
	computer accessories	6.7
	Furniture Decoration	5.4
	housewares	4.7
	Cool Stuff	4.7
	automotive	4.4
	Garden tools	3.6

Q4. Correlation Between Product Price and Purchase Count.

Python Code :

```
plot_df['label'] = plot_df['product_id'].astype(str).str[:5]

fig, ax1 = plt.subplots(figsize=(12, 7))

bars = ax1.bar(plot_df['label'], plot_df['purchase_count'], color='coral', alpha=0.7)
ax1.set_ylabel('Purchase Count', color='indianred', fontsize=12)
ax1.tick_params(axis='y', labelcolor='indianred')

ax1.bar_label(bars, padding=3)

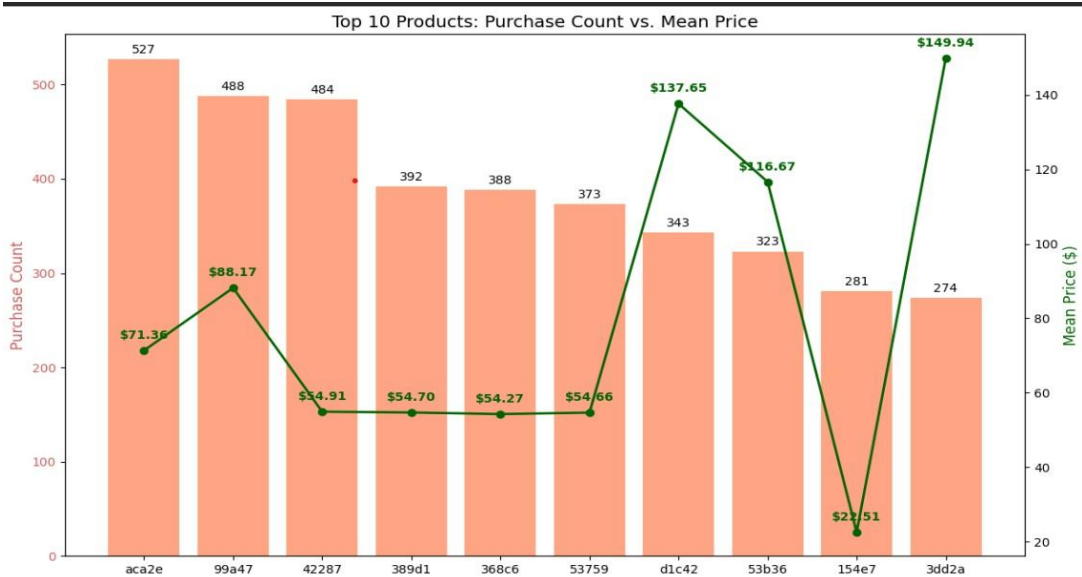
ax2 = ax1.twinx()
ax2.plot(plot_df['label'], plot_df['mean_price'], color='darkgreen', marker='o', linewidth=2)
ax2.set_ylabel('Mean Price ($)', color='darkgreen', fontsize=12)
ax2.tick_params(axis='y', labelcolor='black')

for i, txt in enumerate(plot_df['mean_price']):
    ax2.annotate(f'${txt:.2f}', (plot_df['label'][i], plot_df['mean_price'][i]),
                textcoords="offset points", xytext=(0,10), ha='center', color='darkgreen', fontwe

# 4. Final Formatting
plt.title('Top 10 Products: Purchase Count vs. Mean Price', fontsize=14)
plt.tight_layout()
plt.show()
```

SQL Query:

```
SELECT
    p.product_id,
    p.product_category_name,
    AVG(oi.price) AS mean_price, -- Matches 'price': 'mean'
    COUNT(oi.order_id) AS purchase_count -- Matches 'order_id': 'count'
FROM products p
JOIN order_items oi ON p.product_id = oi.product_id
GROUP BY p.product_id, p.product_category_name
ORDER BY purchase_count DESC
limit 10;
```



	product_id	product_category_name	mean_price	purchase_count
▶	aca2eb7d00ea1a7b8ebd4e68314663af	Furniture Decoration	71.364137	527
	99a4788cb24856965c36a24e339b6058	bed table bath	88.167131	488
	422879e10f46682990de24d770e7f83d	Garden tools	54.911612	484
	389d119b48cf3043d311335e499d9c6b	Garden tools	54.695383	392
	368c6c730842d78016ad823897a372db	Garden tools	54.270103	388
	53759a2eccdad2bb87a079a1f1519f73	Garden tools	54.657373	373
	d1c427060a0f73f6b889a5c7c61f2ac4	computer accessories	137.651633	343
	53b36df67ebb7c41585e8d54d6772e08	Watches present	116.666935	323
	154e7e31ebfa092203795c972e5804a6	HEALTH BEAUTY	22.509573	281
	3dd2a17168ec895c781a9191c1e95ad7	computer accessories	149.936496	274

Q5. Total Revenue Generated by Each Seller (Ranked).

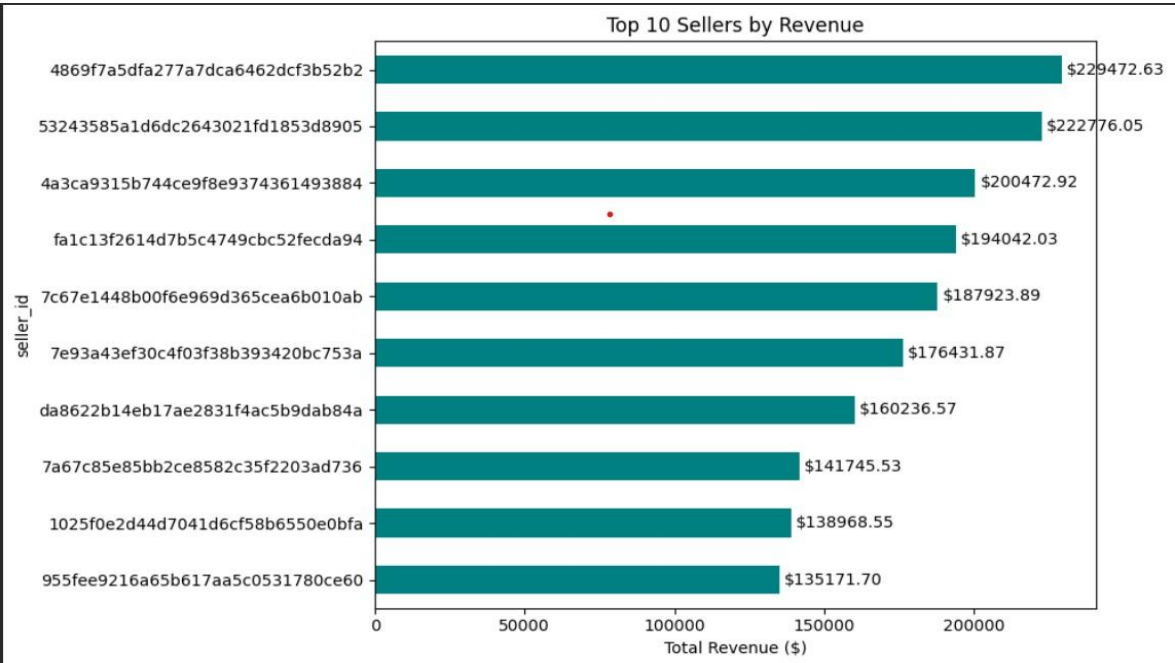
Python Code :

```
seller_rev = order_items.groupby('seller_id')['price'].sum().sort_values(ascending=False).head(10)

plt.figure(figsize=(10, 6))
ax = seller_rev.plot(kind='barh', color='teal')

ax.bar_label(ax.containers[0], fmt='%.2f', padding=3)

plt.title('Top 10 Sellers by Revenue')
plt.xlabel('Total Revenue ($)')
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```



SQL Query:

```
SELECT
    seller_id,
    SUM(price) AS total_revenue,
    DENSE_RANK() OVER (ORDER BY SUM(price) DESC) AS seller_rank
FROM order_items
GROUP BY seller_id
limit 10;
```

	seller_id	total_revenue	seller_rank
▶	4869f7a5dfa277a7dca6462dcf3b52b2	229472.63	1
	53243585a1d6dc2643021fd1853d8905	222776.05	2
	4a3ca9315b744ce9f8e9374361493884	200472.92	3
	fa1c13f2614d7b5c4749cbc52fecda94	194042.03	4
	7c67e1448b00f6e969d365cea6b010ab	187923.89	5
	7e93a43ef30c4f03f38b393420bc753a	176431.87	6
	da8622b14eb17ae2831f4ac5b9dab84a	160236.57	7
	7a67c85e85bb2ce8582c35f2203ad736	141745.53	8
	1025f0e2d44d7041d6cf58b6550e0bfa	138968.55	9
	955fee9216a65b617aa5c0531780ce60	135171.70	10

Advanced Level Question

Q1. Moving Average of Order Values.

Python Code :

```
df = pd.merge(orders, payments, on='order_id')

df['order_purchase_timestamp'] = pd.to_datetime(df['order_purchase_timestamp'])
df = df.sort_values(['customer_id', 'order_purchase_timestamp'])

df['moving_avg'] = (df.groupby('customer_id')['payment_value']
                    .expanding()
                    .mean()
                    .reset_index(level=0, drop=True))

result = df[['moving_avg']]
print(result.head(10))
```

SQL Query:

```
SELECT
    customer_id,
    order_purchase_timestamp,
    payment_value,
    AVG(payment_value) OVER (PARTITION BY customer_id ORDER BY order_purchase_timestamp) AS moving_avg
FROM orders o
JOIN payments p ON o.order_id = p.order_id
limit 10;
```

```
...      moving_avg
71587      114.74
10466       67.41
68795      195.42
45159      179.35
6119       107.01
76895       71.80
48295       49.40
62657      166.59
82776       85.23
83947     1255.71
```

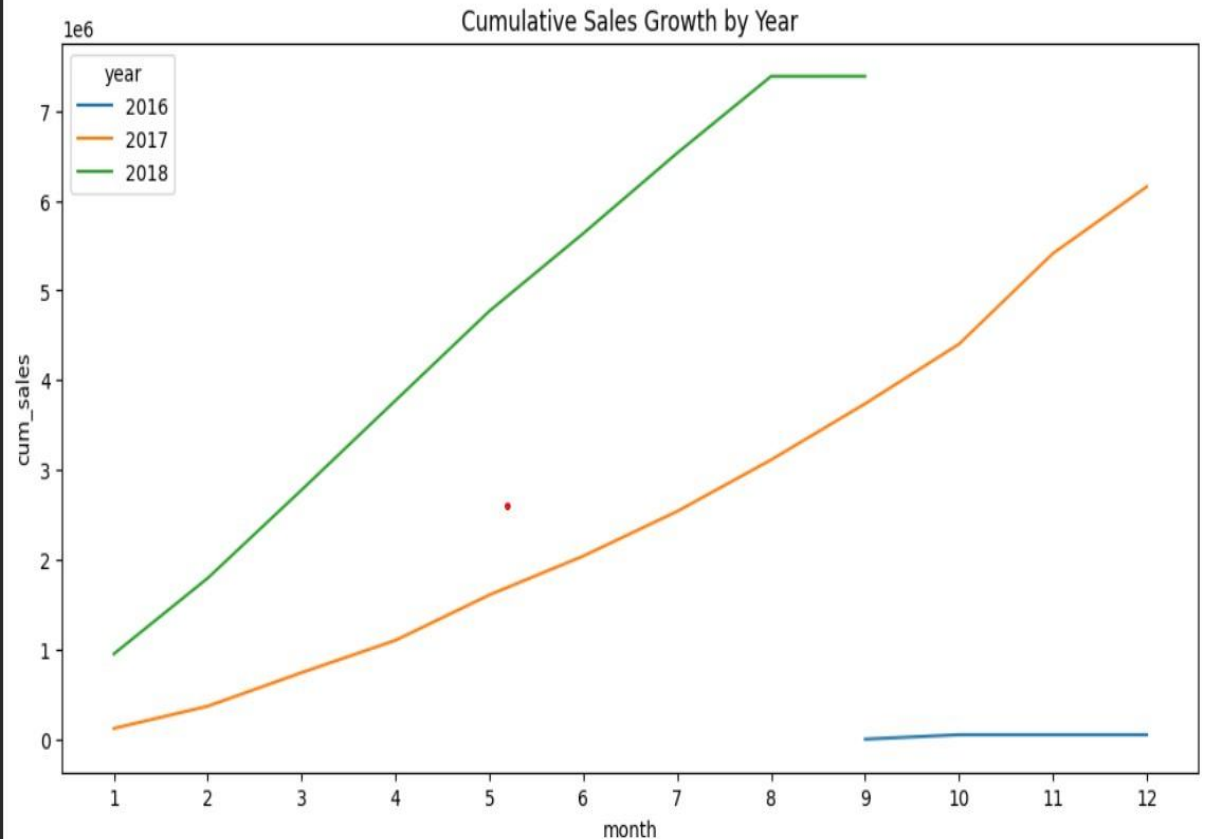
	customer_id	order_purchase_timestamp	payment_value	moving_avg
▶	00012a2ce6f8dcda20d059ce98491703	2017-11-14 16:08:26	114.74	114.740000
	000161a058600d5901f007fab4c27140	2017-07-16 09:40:32	67.41	67.410000
	0001fd6190edaaf884bcdf3d49edf079	2017-02-28 11:06:43	195.42	195.420000
	0002414f95344307404f0ace7a26f1d5	2017-08-16 13:09:20	179.35	179.350000
	000379cdec625522490c315e70c7a9fb	2018-04-02 13:42:17	107.01	107.010000
	0004164d20a9e969af783496f3408652	2017-04-12 08:35:12	71.80	71.800000
	000419c5494106c306a97b5635748086	2018-03-02 17:47:40	49.40	49.400000
	00046a560d407e99b969756e0b10f282	2017-12-18 11:08:30	166.59	166.590000
	00050bf6e01e69d5c0fd612f1bcfb69c	2017-09-17 16:04:44	85.23	85.230000
	000598caf2ef4117407665ac33275130	2018-08-11 12:14:35	1255.71	1255.710000

Q2. Cumulative Sales per Month for Each Year.

Python Code :

```
order_val = order_items.groupby('order_id')['price'].sum().reset_index()
sales_trend = pd.merge(orders[['order_id', 'order_purchase_timestamp']], order_val, on='order_id')
sales_trend['year'] = sales_trend['order_purchase_timestamp'].dt.year
sales_trend['month'] = sales_trend['order_purchase_timestamp'].dt.month
monthly_sales = sales_trend.groupby(['year', 'month'])['price'].sum().reset_index()
monthly_sales['cum_sales'] = monthly_sales.groupby('year')['price'].cumsum()

plt.figure(figsize=(12, 6))
sns.lineplot(data=monthly_sales, x='month', y='cum_sales', hue='year', palette='tab10')
plt.title('Cumulative Sales Growth by Year')
plt.xticks(range(1, 13))
plt.show()
```



SQL Query:

-- CUMULATIVE SALES BY MONTH AND YEAR

```
WITH MonthlySales AS (  
    SELECT  
        YEAR(o.order_purchase_timestamp) AS sales_year,  
        MONTH(o.order_purchase_timestamp) AS sales_month,  
        SUM(p.payment_value) AS monthly_revenue  
    FROM orders o  
    JOIN payments p ON o.order_id = p.order_id  
    GROUP BY sales_year, sales_month  
)  
SELECT  
    sales_year, sales_month, monthly_revenue,  
    SUM(monthly_revenue) OVER (PARTITION BY sales_year ORDER BY sales_month) AS cumulative_sales  
FROM MonthlySales;
```

	sales_year	sales_month	monthly_revenue	cumulative_sales
▶	2016	9	252.24	252.24
	2016	10	59090.48	59342.72
	2016	12	19.62	59362.34
	2017	1	138488.04	138488.04
	2017	2	291908.01	430396.05
	2017	3	449863.60	880259.65
	2017	4	417788.03	1298047.68
	2017	5	592918.82	1890966.50
	2017	6	511276.38	2402242.88
	2017	7	592382.92	2994625.80
	2017	8	674396.32	3669022.12
	2017	9	727762.45	4396784.57
	2017	10	779677.88	5176462.45
	2017	11	1194882.80	6371345.25
	2017	12	878401.48	7249746.73
	2018	1	1115004.18	1115004.18
	2018	2	992463.34	2107467.52
	2018	3	1159652.12	3267119.64
	2018	4	1160785.48	4427905.12
	2018	5	1153982.15	5581887.27
	2018	6	1023880.50	6605767.77
	2018	7	1066540.75	7672308.52
	2018	8	1022425.32	8694733.84
	2018	9	4439.54	8699173.38
	2018	10	589.67	8699763.05

Q3. Year-over-Year (YoY) Growth Rate.

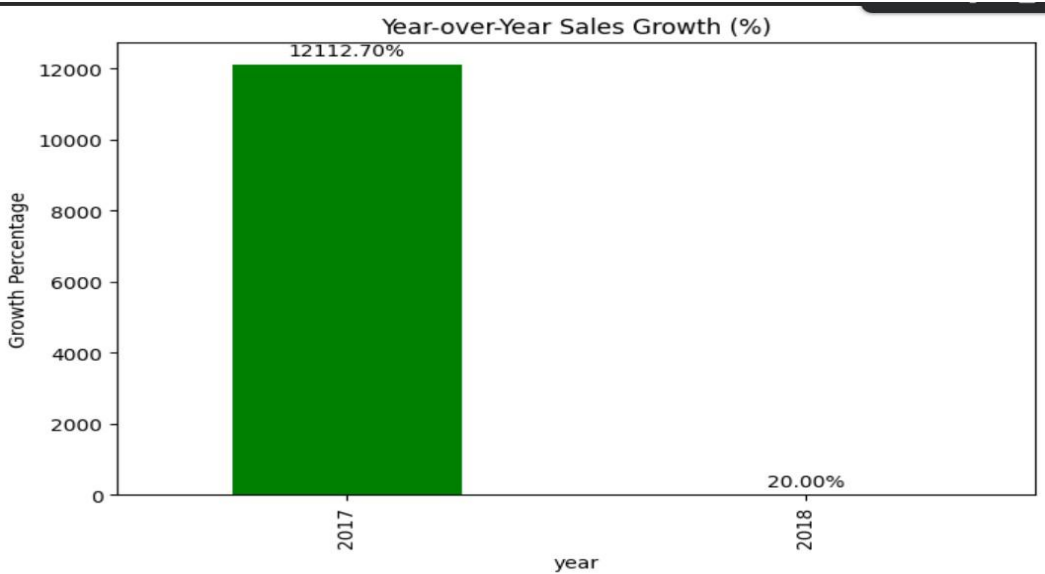
Python Code :

```
df4 = pd.merge(orders, payments, on='order_id')

df4['year'] = pd.to_datetime(df4['order_purchase_timestamp']).dt.year
yearly_sales = df4.groupby('year')['payment_value'].sum()

yoy_growth = yearly_sales.pct_change() * 100

ax = yoy_growth.dropna().plot(kind='bar', figsize=(8, 5), color='green')
ax.bar_label(ax.containers[0], fmt='%.2f%', padding=3)
plt.title('Year-over-Year Sales Growth (%)')
plt.ylabel('Growth Percentage')
plt.show()
```



SQL Query:

```
WITH YearlySales AS (
    SELECT
        YEAR(order_purchase_timestamp) AS year,
        SUM(payment_value) AS total_revenue
    FROM orders o
    JOIN payments p ON o.order_id = p.order_id
    GROUP BY year
)
SELECT
    year,
    total_revenue,
    LAG(total_revenue) OVER (ORDER BY year) AS previous_year_revenue,
    ((total_revenue - LAG(total_revenue) OVER (ORDER BY year)) / LAG(total_revenue) OVER (ORDER BY year)) * 100 AS yoy_growth
FROM YearlySales;
```

	year	total_revenue	previous_year_revenue	yoy_growth
▶	2016	59362.34	NULL	NULL
	2017	7249746.73	59362.34	12112.703761
	2018	8699763.05	7249746.73	20.000924

Q4. Retention Rate (Repeat Purchase within 6 Months).

Python Code :

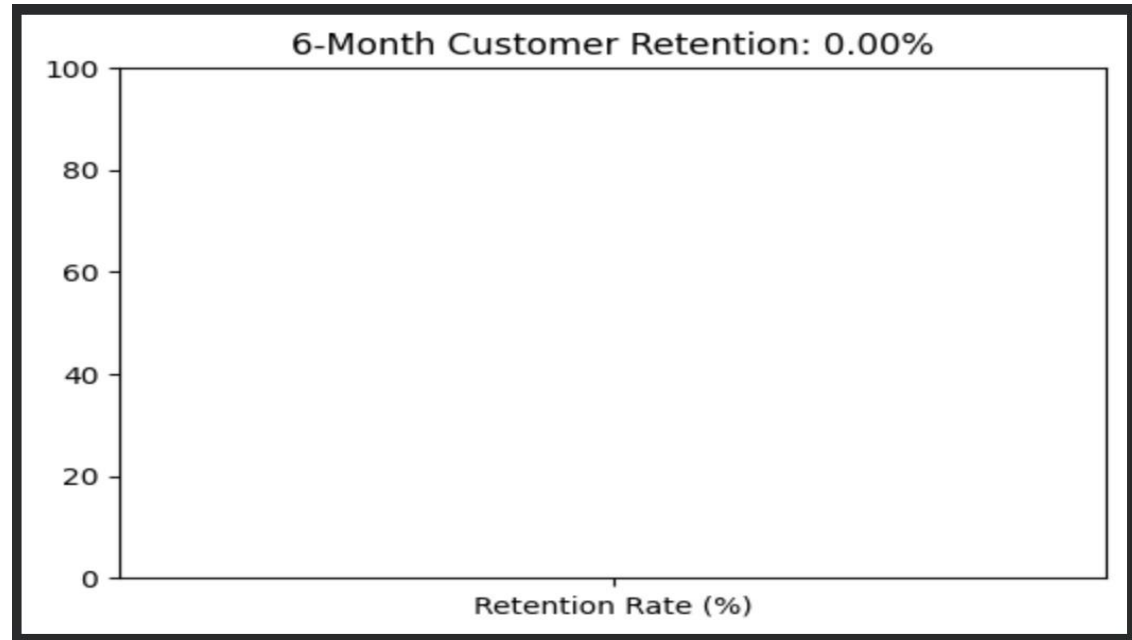
```
first_purchase = orders.groupby('customer_id')['order_purchase_timestamp'].min().reset_index()
first_purchase.columns = ['customer_id', 'first_order_date']

retention_data = pd.merge(orders[['customer_id', 'order_purchase_timestamp']], first_purchase, on=
retention_data['days_diff'] = (retention_data['order_purchase_timestamp'] - retention_data['first_

retained_customers = retention_data[(retention_data['days_diff'] > 0) & (retention_data['days_diff
total_unique_customers = customers['customer_id'].nunique()

retention_rate = (retained_customers / total_unique_customers) * 100

plt.figure(figsize=(6, 4))
plt.bar(['Retention Rate (%)'], [retention_rate], color='green')
plt.ylim(0, 100)
plt.title(f'6-Month Customer Retention: {retention_rate:.2f}%')
plt.show()
```



SQL Query:

```
WITH CustomerStats AS (
    SELECT
        customer_id,
        MIN(order_purchase_timestamp) AS first_order,
        MAX(order_purchase_timestamp) AS last_order,
        COUNT(order_id) AS total_orders
    FROM orders
    GROUP BY customer_id
)
SELECT
    (COUNT(CASE WHEN total_orders > 1
        AND last_order <= DATE_ADD(first_order, INTERVAL 6 MONTH)
        THEN 1 END) / COUNT(*)) * 100 AS retention_rate
FROM CustomerStats;
```

	retention_rate
▶	0.0000

Q5. Top 3 Customers Who Spent the Most Each Year.

Python Code :

```
df4 = pd.merge(orders, payments, on='order_id')
df4['year'] = df4['order_purchase_timestamp'].dt.year

top_3_per_year_raw = df4.groupby(['year', 'customer_id'])['payment_value'].sum().reset_index()
top_3_per_year = top_3_per_year_raw.sort_values(by=['year', 'payment_value'], ascending=[True, False])

top_3_per_year['cust_label'] = top_3_per_year['customer_id'].astype(str).str[:8]

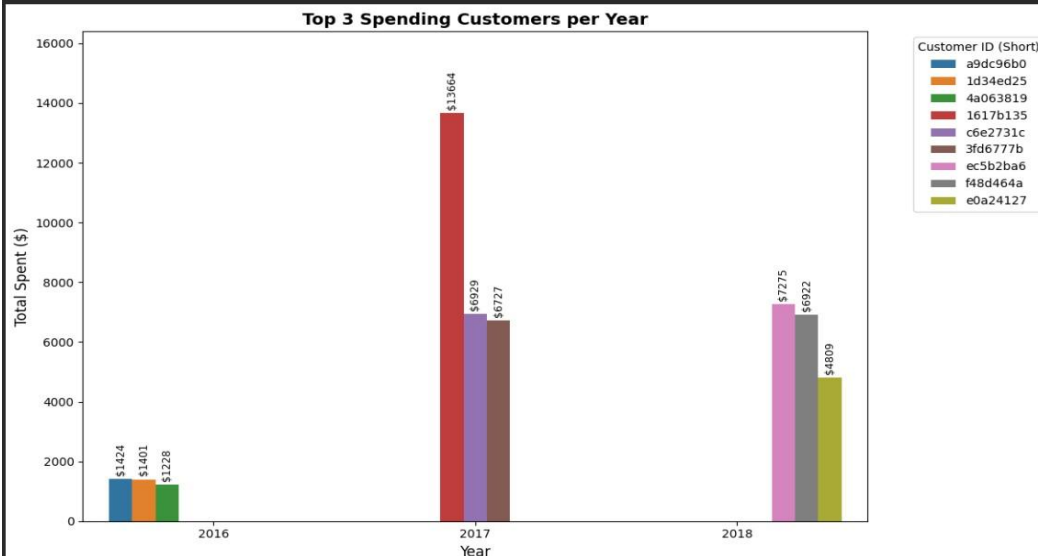
plt.figure(figsize=(12, 7))
ax = sns.barplot(data=top_3_per_year, x='year', y='payment_value', hue='cust_label')

for container in ax.containers:
    ax.bar_label(container, fmt='${:.0f}', padding=3, fontsize=9, rotation=90)

plt.title('Top 3 Spending Customers per Year', fontsize=14, fontweight='bold')
plt.xlabel('Year', fontsize=12)
plt.ylabel('Total Spent ($)', fontsize=12)

plt.margins(y=0.2)

plt.legend(title='Customer ID (Short)', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```



SQL Query:

```
WITH CustomerSpending AS (
    SELECT
        YEAR(o.order_purchase_timestamp) AS year,
        o.customer_id,
        SUM(p.payment_value) AS total_spent
    FROM orders o
    JOIN payments p ON o.order_id = p.order_id
    GROUP BY year, o.customer_id
),
RankedCustomers AS (
    SELECT
        year, customer_id, total_spent,
        RANK() OVER (PARTITION BY year ORDER BY total_spent DESC) AS spend_rank
    FROM CustomerSpending
)
SELECT * FROM RankedCustomers WHERE spend_rank <= 3;
```

	year	customer_id	total_spent	spend_rank
▶	2016	a9dc96b027d1252bbac0a9b72d837fc6	1423.55	1
	2016	1d34ed25963d5aae4cf3d7f3a4cda173	1400.74	2
	2016	4a06381959b6670756de02e07b83815f	1227.78	3
	2017	1617b1357756262bfa56ab541c47bc16	13664.08	1
	2017	c6e2731c5b391845f6800c97401a43a9	6929.31	2
	2017	3fd6777bbce08a352fddd04e4a7cc8f6	6726.66	3
	2018	ec5b2ba62e574342386871631fafd3fc	7274.88	1
	2018	f48d464a0baaea338cb25f816991ab1f	6922.21	2
	2018	e0a2412720e9ea4f26c1ac985f6a7358	4809.44	3

KEY INSIGHT

- A big portion of the customer base is concentrated in specific states, with **São Paulo (SP)** leading by a wide margin (41,746 customers), followed by **Rio de Janeiro (RJ)** and **Minas Gerais (MG)**.
- The top revenue-generating product categories are **Health & Beauty** (\$1,297,355.80) and **Watches/Presents** (\$1,253,143.30).
- There is a nearly equal split between customers paying in a **single instalment (50.6%)** versus those using **multiple instalments (49.4%)**.
- In 2017, the platform recorded **45,101 orders**. The 2018 monthly trend shows peak order volumes in **May (7,269 orders)** and **August (7,211 orders)**.
- Customers are spread across a diverse range of **4,119 unique cities**.

RECOMMENDATIONS

- Since São Paulo, Rio de Janeiro, and Minas Gerais represent the largest customer hubs, focus digital advertising and promotional campaigns in these regions to maximize ROI.
- Prioritize stock and logistics for high demand categories like **Health & Beauty** and **Watches/Presents**, as these are the primary revenue drivers.
- Given that nearly half of the customers prefer instalments, ensure that payment gateways remain seamless and consider offering "interest-free" instalment promotions to further incentivize high-value purchases.
- Align major sales events or product launches with observed peak months like **May and August** to capitalize on established high-traffic periods.
- While the customer base is broad (4,119 cities), the steep drop-off in customer counts outside the top states suggests an opportunity to investigate logistics or marketing barriers in underrepresented regions to grow the market share.